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Journal of Environmental Planning and Management

ISSN: 0964-0568 (Print) 1360-0559 (Online) Journal homepage: www.tandfonline.com/journals/cjep20

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To cite this article: Bon Woo Koo, Donghwan Ki, Youjung Kim, Chaeyeon Han & Sungmin Lee (07 Sep 2026): Leveraging AI for plan evaluation: assessing ChatGPT's potential and limitations for a flood resilience scorecard, Journal of Environmental Planning and Management, DOI: 10.1080/09640568.2026.2714014

To link to this article: https://doi.org/10.1080/09640568.2026.2714014

Published online: 07 Sep 2026.

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Journal of Environmental Planning and Management, 2026
https://doi.org/10.1080/09640568.2026.2714014

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Leveraging AI for plan evaluation: assessing ChatGPT's potential and limitations for a flood resilience scorecard

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(*Received 19 February 2025; final version received 13 July 2026*)

Plan evaluation is essential for improving plan quality and fostering resilient development amid climate change. Traditionally, plan evaluation required significant time and resources, making timely responses to emerging challenges difficult. This study explores the potential of ChatGPT for accurately automating plan evaluation by examining its performance compared to human examiners. Using the Plan Integration for Resilience Scorecard™, the study evaluated the Imagine 2040: Tampa Comprehensive Plan. Findings showed ChatGPT effectively identified policies relevant to flood resilience, but human verification was still necessary. Once relevant policies were identified, ChatGPT achieved a 90% agreement rate with human evaluators in assessing whether policies would increase or decrease flood vulnerability. ChatGPT required significantly less time compared to human examiners. ChatGPT can serve as a complementary tool for rapid plan evaluation with human oversight. This study highlights implications for policymakers and practitioners, discusses caveats of using ChatGPT in plan evaluation, and identifies future research directions.

Keywords: plan evaluation; flood resilience; resilience scorecard; artificial intelligence; ChatGPT

1. Introduction

Plans and policies are blueprints of the future development trajectories of cities, guiding decisions on land use, infrastructure development, and public services while balancing economic growth with social equity and environmental sustainability (Berke *et al*. 2006). High-quality plans are therefore more likely to support improved future outcomes and effective implementation (Guyadeen 2018). Plan evaluation is an indispensable analytical practice in this regard, providing a systematic framework for assessing and improving plan quality (Berke and Godschalk 2009). As planning contexts grow more dynamic and uncertain – driven by the accelerating pressures of climate change and other rapid urban transformations – the ability to perform accurate and timely plan evaluation has become increasingly important (Mimura *et al*. 2015).

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Flooding is one of the climate change-induced hazards that is increasing in frequency and intensity, posing critical risks to many cities around the world (Dodman *et al*. 2022; Wing *et al*. 2022). Accordingly, many plan quality evaluations have focused on hazard mitigation, coastal planning, watershed protection, and land use (Berke, Smith, and Lyles 2012; Berke *et al*. 2013, Berke *et al*. 2019; Malecha, Woodruff, and Berke 2021). In the context of flood-related resilience, rigorous and timely plan evaluation can help determine whether the existing and planned strategies adequately protect communities, mitigate risks, and support recovery. Yet the capacity to conduct such evaluation at the scale and speed demanded by a changing climate and other rapid urban transformations remains a persistent challenge.

Traditional methods of plan evaluation typically rely on intensive human labor as they commonly involve a combination of qualitative and quantitative approaches, such as surveys, interviews and workshops, case studies, and statistical analysis, to assess both internal coherence and external effectiveness of plans (American Planning Association 2012; Berke *et al*. 2006; Berke and Godschalk 2009; Brody and Highfield 2005; Guyadeen and Seasons 2016). While these traditional methods provide rigorous and contextually rich insights, they often pose several challenges: inconsistency in judgment, the need for specialized expertise, and the high costs of reviewing the growing volume of planning documents (Bell 2004; Berke *et al*. 2019; Guyadeen and Seasons 2016; Stevens, Lyles, and Berke 2014). These constraints raise a practical question: can computational tools – particularly modern large language models (LLMs) like ChatGPT – assist in making plan evaluations faster, more consistent, and more scalable without sacrificing analytical rigor?

Since the introduction of ChatGPT and other similar LLMs in recent years, numerous studies have highlighted their potential to support – and, in some cases, even automate – human labor in tasks traditionally considered exclusive to human intelligence (Biswas 2023; Cascella *et al*. 2023; Herbold *et al*. 2023; Katz *et al*. 2024). ChatGPT has demonstrated its ability to analyze large volumes of text data, identify thematic patterns, and provide real-time insights (Fu, Wang, and Li 2024; Thelwall 2025). Applied to plan evaluation, such capabilities could enable policymakers to rapidly assess the extent to which plans support or hinder communities' resilience to flooding at scale – complementing established human-led methods and producing more timely and comprehensive policy insights.

This study investigates whether ChatGPT can serve as a reliable and accurate tool for assisting plan evaluation focused on resilience planning by systematically comparing its evaluation results with those of trained human examiners. Specifically, we examine how different configurations, including distinct personas and prompt designs, affect evaluation results at different stages of plan evaluation process, and assess response consistency across repeated trials as a proxy for reliability and robustness from hallucinations. The findings contribute empirical evidence and practical guidance for integrating LLMs into plan evaluation workflows, with particular relevance to flood resilience planning under climate uncertainty.

2. Literature review

2.1. Role of plan evaluation tools in resilience planning

Policies and plans are the core tools through which urban planners shape the future of cities. Plan evaluation refers to the systematic assessment of planning processes and

outcomes using clearly defined criteria (Laurian *et al*. 2010), and examines the quality, coherence, and intended impacts of long-range policy documents that guide future development (Berke *et al*. 2006; Guyadeen and Seasons 2016). The evaluation process plays a critical role in ensuring that policies and plans are aligned with the objectives of enhancing resilience and fostering sustainable urban development (Berke and Godschalk 2009; S. C. Woodruff *et al*. 2022). Since the 1990s, planning researchers have developed various frameworks for evaluating plan quality, providing standardized tools for assessing planning processes and outcomes across jurisdictions (Guyadeen 2018; Stevens, Lyles, and Berke 2014).

One application of plan evaluation in resilience planning is identifying inconsistencies across multiple plans. The increasing frequency and severity of natural hazards driven by climate change further necessitate comprehensive and integrated approaches to resilience planning (IPCC 2014). However, local plans—such as comprehensive, transportation, hazard mitigation, housing, and other plans governing land use and development – often contain conflicting policies in hazard-prone areas (Berke *et al*. 2015; Malecha *et al*. 2019). For example, Berke *et al*. (2015) used a plan evaluation tool – the Plan Integration for Resilience Scorecard (PIRS™) – to identify that the Coastal Area Management Act (CAMA) Land Use Plan in the City of Washington, NC, contained policies that increased physical vulnerability in the 100-year floodplain, while the local hazard mitigation plan contained policies aimed at reducing it. Similarly, Malecha *et al*. (2019) demonstrated that the comprehensive plan in League City, TX, supported increased development in the 100-year floodplain, while the hazard mitigation plan incorporated a land acquisition strategy for flood-prone properties – yet the two were not coordinated. Such conflicts can undermine the effectiveness of vulnerability-reducing policies or may even exacerbate vulnerability to flood hazards.

Despite the utility of these plan evaluation tools, the resource-intensiveness of the plan evaluation process makes timely application challenging for many municipalities (DeAngelis *et al*. 2021). Conducting a thorough evaluation can take months and involve multiple individuals with specialized expertise and local knowledge – a particularly burdensome undertaking in resource-constrained communities (Cusick 2021; Hughes, Resor, and Newberry 2023). This challenge is compounded by the growing number and volume of planning documents, as municipalities increasingly develop or update climate change and resilience plans (Fu, Wang, and Li 2024; S. Woodruff *et al*. 2021).

2.2. Artificial intelligence (AI) for plan evaluation: promise and limitations

In response to the resource-intensiveness of plan evaluation, the advances in AI, and rapidly evolving challenges such as climate change, the integration of AI into public governance has significantly expanded, particularly as the COVID-19 pandemic accelerated the digital transformation of government operations (Jiang *et al*. 2023). In the plan evaluation domain, a prominent application of AI is Natural Language Processing (NLP), a subfield of AI that facilitates the analysis of large textual data, such as government reports, policy documents, and social media content. NLP techniques such as text classification, sentiment analysis, and topic modeling automate the extraction of relevant information, enhancing the efficiency of plan evaluations. For instance, these tools enable real-time monitoring of public sentiment and policy impacts, which are vital in fields like healthcare, economic planning, and disaster management (Sukhwal

and Kankanhalli 2022). During the COVID-19 pandemic, NLP was instrumental in analyzing public sentiment and behavioral shifts in response to health policies, providing policy insights that informed public health strategies (Gupta *et al*. 2021; Misra *et al*. 2018; Sukhwal and Kankanhalli 2022). Furthermore, text classification and topic modeling enable categorization and summarization of vast amounts of policy-related documents (Gunes and Florczak 2023), improving accessibility and usability for government officials.

The emergence of LLMs such as ChatGPT, Gemini, Claude, and Llama, has expanded the potential of AI in plan evaluation even further. These models offer considerable strengths, including the ability to generate coherent text and synthesize large volumes of information into concise summaries in near real time (D. Wang, Lu, and Fu 2023). Additionally, LLMs significantly lower the barriers to adopting computational text analysis by eliminating the need for extensive domain expertise, large training corpora, and manual preprocessing of text data, while offering ready-to-use solutions that simplify implementation and enhance reproducibility (Gunes and Florczak 2023). Such features improve accessibility to policy details for non-expert audiences, potentially broadening awareness and engagement.

While LLMs have shown impressive performances in tasks that have traditionally been viewed as exclusive to human intelligence, LLMs also present notable limitations. A significant concern lies in the potential for biases embedded in the training data, which primarily originate from online sources and may lack the geographical and social specificity required for plan evaluation (J. Kim, Lee, *et al*. 2024). Moreover, the performance of LLMs is highly dependent on the specificity of input prompts, which can lead to inconsistencies (J. H. Lee and Shin 2024; Murr, Grainger, and Gao 2023). This non-deterministic nature becomes particularly problematic in high-context scenarios, such as evaluating policies with localized or culturally sensitive implications. Another significant limitation is the "hallucination effect," where LLMs generate information that appears credible but is factually fabricated. This issue is especially relevant when evaluating policies that require accurate and context-specific information, as misinformation can undermine the reliability of the analysis and lead to flawed conclusions. Addressing these challenges is critical to ensure their outputs remain reliable and contextually appropriate (Atkins *et al*. 2024; Shen *et al*. 2023).

These advantages and disadvantages of LLMs are empirically documented in several research papers that employ ChatGPT as a policy assessment tool. For instance, Tang *et al*. (2023) evaluated the capability of 'PolicyGPT' to categorize privacy policy segments based on datasets annotated by legal experts. Under zero-shot learning conditions, PolicyGPT achieved accuracy rates of 97% and 87% across two datasets, surpassing traditional machine learning models. This underscores its utility in efficiently analyzing verbose documents like privacy policies, which often consist of large volumes of text full of legal jargon. In the urban planning domain, Fu, Wang, and Li (2024) explored ChatGPT's application in evaluating climate change plans. They reviewed 10 climate change plans from the USA by generating a series of binary results per plan regarding resilience-related components, and compared the machine-generated results with those from human coders. Their findings revealed that the AI tool could approximate human evaluations in about 68% of cases, identifying overlooked elements and offering consistent evaluations. These findings underscore its

potential in tasks that involve processing large textual datasets, though the study highlighted the challenges of domain-specific jargon and context sensitivity.

The reviewed studies collectively suggest that while ChatGPT and similar AI tools are not yet capable of replacing human expertise, they are effective complements. Notably, these AI tools excel at managing tasks that traditionally required tedious human input, significantly enhancing overall efficiency. These studies suggest that AI integration into plan evaluation can improve efficiency and reduce human errors, particularly in tasks involving large-scale document analysis.

2.3. Research gap

In summary, while plan evaluation is an essential process for ensuring that plans and policies effectively align with goals such as resilience, equity, and sustainability, traditional methods face several limitations. These include the labor-intensive nature of evaluation tools, dependence on human expertise, and the challenge of reviewing the vast volume of documents. Furthermore, existing studies that apply LLMs to plan evaluation typically focus on special-purpose plans and policies with narrowly defined domains, such as privacy policies and climate change plans (Fu, Wang, and Li 2024; Tang *et al*. 2023). This narrow focus limits the generalizability of their findings to resilience planning because the majority of policies relevant to flood resilience are embedded within comprehensive plans – which cover a wide range of topics, many of which have little relevance to flood resilience (Berke *et al*. 2006; Y. Kim, Newman, *et al*. 2024). This mixed nature means that applying LLM-based evaluation to these plans requires first determining whether a provision is relevant at all—a task that previous research has not examined – before assessing how it impacts resilience.

To address these gaps, this study seeks to assess the applicability of ChatGPT in plan evaluation by examining its accuracy at each stage of the process. This investigation can be further divided into the following two research questions:

1. Research Question 1: How accurately and efficiently can ChatGPT identify policies or policy-like statements relevant to flood resilience from a comprehensive plan that covers a broad range of policy domains beyond flood resilience?
2. Research Question 2: Once the relevant policies are identified, how accurately does ChatGPT evaluate the directionality of those policies' impacts (e.g., whether the impact would be positive or negative) on vulnerability to flooding?

By answering these questions, this study aims to expand the literature on LLM-assisted plan evaluation by offering a more granular assessment of LLM performance across different stages, narrowing the gap between traditional and AI-assisted approaches, and offering practical lessons for enhancing flood resilience planning through AI integration.

3. Data and methods

3.1. Selection of study site and plan evaluation tool

The plan evaluation method was selected based on three considerations. First, the method should have a specific focus on climate resilience or related topics such as hazard resilience. Second, the selected method should dissect a plan into granular

components and evaluate them individually, allowing users to pinpoint where the AI falls short of replicating human judgement. Third, the selected method should be accompanied by a detailed official guidebook that provides clearly defined procedural instructions and evaluation criteria. A detailed guidebook establishes a common evaluative basis that both human examiners and ChatGPT can reference, allowing observed differences to be more meaningfully interpreted as reflections of ChatGPT's reasoning rather than artifacts of inconsistent evaluation standards.

We considered various plan evaluation frameworks, including APA Comprehensive Plan Standards for Sustaining Places (Godschalk and Rouse 2015) and the Disaster Resilience Scorecard for Cities (United Nations Office for Disaster Risk Reduction 2017), and selected the PIRS™ as it best meets all three of the above criteria and has shown effectiveness in previous studies (Berke *et al*. 2015; Y. Kim, Newman, *et al*. 2024). PIRS™ is expressly designed to assess hazard mitigation and adaptation strategies within or across plans and examine policy conflicts among them, ensuring domain relevance. Furthermore, its structure – which divides a plan evaluation process into distinct stages and evaluates specific policy items using well-defined evaluation criteria rather than assigning a holistic score – enables the precise identification of where AI falls short, providing a rigorous framework for testing the efficacy of prompt engineering techniques. Finally, it has the official Guidebook (Malecha *et al*. 2019) in which detailed descriptions of processes and evaluation criteria are provided.

The PIRS™ process, as outlined in the official Guidebook (Malecha *et al*. 2019), consists of three key phases: (1) the evaluation of each policy in relevant plans for their impact on resilience (i.e. Prepare Policies section in the guidebook), (2) the identification of the affected areas for each policy (i.e. Prepare Maps section), and (3) the aggregation of the policy evaluation results by the area (i.e. Assign Policy Scores section). The first phase – evaluating the policies for their impact on resilience – requires a manual evaluation of each policy in various plans to determine whether the policies are relevant to flood vulnerability and, if relevant, whether the policies increase or decrease vulnerabilities. In contrast, the second and third phases are primarily data-driven and can be executed at scale through the use of geographic information systems (GIS) and readily available data such as planning district boundaries, National Flood Insurance Program (NFIP) flood maps, and Census data. Therefore, this study focuses on leveraging generative AI for the first phase of the PIRS™ process.

The first phase – evaluating the policies for their impact on resilience – can be subdivided into two analytical steps for clarity: (1) identifying policies relevant to flood vulnerability and (2) evaluating whether those policies are expected to increase or decrease vulnerability. These steps are referred to as Task 1 and Task 2, respectively, throughout this study. The two tasks correspond with the two research questions posed in Section 2.3. First, Task 1 evaluates every individual policy or policy-like statement in a comprehensive plan (e.g. a comprehensive plan of a city) for its relevance to flooding resilience (i.e. Research Question 1). Once the relevant policies or policy-like statements are identified, Task 2 assigns either a score of "+1" (i.e. decreasing vulnerability) or "−1" (i.e. increasing vulnerability) to each relevant policy (i.e. Research Question 2).

This study uses the Imagine 2040: Tampa Comprehensive Plan, updated in 2024 (hereafter referred to as 'the Plan') as a case study. The City of Tampa in Florida is highly vulnerable to coastal and river flooding (City of Tampa 2023; Kim and Newman 2020). Past studies have found that the majority of policies identified as

relevant to flood vulnerability are from comprehensive plans due to their role in offering community visions for future development and broadly covering social, economic, and environmental issues (Berke *et al*. 2019; Y. Kim, Newman, *et al*. 2024). The Plan also contains a large number of policies, making it suitable for evaluating the accuracy and reliability of assessments made by ChatGPT. Thus, we use the Tampa Comprehensive Plan 2040 as a pilot case study to examine ChatGPT's performance in selecting policies and scoring their relevance and directions. We used Python programming language to automatically extract individual policies from a PDF version of the Plan. This list was evaluated independently by human examiners and ChatGPT, respectively, through the steps delineated below. Subsequently, their results were compared to assess the agreement between them.

3.2. Resilience scorecard by human examiners

3.2.1. Task 1 by human examiner: selecting relevant policies

For Task 1, two human examiners independently assessed the policy list, using their professional knowledge of the study area and flood vulnerability, to identify policies relevant to flood vulnerability. The human examiners followed the PIRS^{TM} Guidebook (Malecha *et al*. 2019) to systematically identify relevant policies (i.e. 'relevant'), distinguishing them from those that do not meet the criteria (i.e. 'irrelevant'), based on the following three criteria:

1. Policies that can affect vulnerability to flooding (i.e., Vulnerability Criterion)
2. Policies that contain place-specific term(s) that can be mapped (i.e., Place Criterion)
3. Policies that contain recognizable policy tool(s) in the Guidebook (i.e., Tool Criterion)

These criteria ensure not only the relevance to flood vulnerability but also the spatial precision and the actionability of a given policy. While the Vulnerability Criterion identifies the general relevance of a policy to flood vulnerability, policies meeting Place Criterion allow them to be spatially matched to specific community areas. Since multiple policies with diverging effects may target the same geographic area, mapping these impact zones is a prerequisite for evaluating the cumulative impacts of various planning efforts. Tool Criterion further refines the selection by ensuring that the policy includes recognizable and actionable tools. As noted by Malecha *et al*. (2019), a policy statement lacking specific language or tools is unlikely to be actionable. Thus, applying the Tool Criterion allows evaluators to distinguish substantive, actionable policies from mere declarations. Then, the two human examiners' results were compared and consolidated by revisiting and resolving any disagreements to arrive at a final outcome.

Example policies and our evaluation results are shown in Table 1. A policy in the Tampa Comprehensive Plan stating, "Provide a link between the various activity centers in the CBD by creating a system of attractive ground-level pedestrian connections," was marked as irrelevant because, although it is actionable and location-specific, it is not related to flood vulnerability. In contrast, the policy, "Encourage economic development/retail opportunities and incentives for the downtown and Channel District CRA districts and surrounding areas," was marked as relevant because it is

Table 1. Examples of human-labeled policy relevance and scoring based on the resilience scorecard methodology.

Policy	Task 1 (identifying relevant policies)	Task 2 (scoring)	Justification
LU Policy 3.2.2: Provide a link between the various activity centers in the CBD by creating a system of attractive ground level pedestrian connections.	Irrelevant	Not applicable	Actionable and place-specific, but not related to flood vulnerability
CM Policy 1.2.4: Encourage public education through advertisements in the media other communication devices on the location of public shelters that are accessible to the residents of Coastal Planning Area.	Irrelevant	Not applicable	Actionable and related to flood vulnerability, but not place-specific
ENV Policy 2.3.4: Protect or enhance Wildlife Corridors by minimizing road crossings over wetlands and/or other surface waters and floodplains and designing crossings for unimpeded wildlife passage provided under local rules and regulations, including mitigation as required.	Relevant	+1	Helps restrain floodplain development as Wildlife Corridors, wetlands, and surface waters are overlapped with the 100-year floodplain zones
CM Policy 1.3.4: Any structure within the 100-Year Floodplain that is damaged in excess of the limits established by FEMA's definition of substantial damage (50% rule) shall be rebuilt to meet or exceed all current building code requirements, including those enacted since the construction of the structure.	Relevant	+1	Ensures buildings within the 100-year floodplain zones comply with flood-specific building code requirements to minimize flood damage.
LU Policy 9.9.5: Encourage economic development/retail opportunities and incentives for the downtown and Channel District CRA districts and surrounding areas.	Relevant	−1	Encourages development with incentives in the area at risk of flooding
HSG Policy 1.1.5: Provide incentives such as development density bonuses, expedited permitting for affordable housing, and eco-friendly green sustainable building standards to encourage developers to include affordable housing in large scale residential projects.	Relevant	−1	Encourages development with density bonus and expedited permits in the area at risk of flooding

actionable and location-specific, and it could affect flood vulnerability by encouraging floodplain development.

3.2.2. Task 2 by human examiner: scoring the selected policies as increasing/decreasing flood vulnerability

Next, human examiners performed Task 2 independently by scoring the directionality of the policies identified as relevant – whether each would increase or decrease vulnerability to flooding – and then comparing their results to determine the final scores. This study evaluated policies assuming they would be applied to areas prone to flooding and identified their directionality. A dichotomous scoring system is used to evaluate policies' anticipated impact on vulnerability to flooding (Berke *et al*. 2015; Malecha *et al*. 2019). Specifically, we evaluated whether a policy is likely to increase (i.e. policy score of "−1") or decrease (i.e. policy score of "+1") vulnerability to flooding.

As shown in Table 1, a policy in the Tampa Comprehensive Plan that states "Protect or enhance Wildlife Corridors by minimizing road crossings over wetlands and/or other surface waters and floodplains" receives a score of "+1." Under the assumption that the policy is applied to areas prone to flooding, this policy would decrease vulnerability to flood by helping minimize development within Wildlife Corridors, wetlands, and/or other surface waters and floodplains. Additionally, protecting natural lands slows surface runoff and increases groundwater holding capacity. On the other hand, another policy that "Provides incentives such as development density bonuses, expedited permitting for affordable housing," can encourage development and therefore increase vulnerability to flood if applied to flood-prone areas, resulting in a score of "−1." The resulting spreadsheet of policies served as the 'ground truth' when compared with the labels from ChatGPT. This ground truth consists of two components: The first is a list of 'relevant' policies and the rest of the policies that are deemed irrelevant. The second component contains a binary policy score ("+1" and "−1") given to the 'relevant' policies.

3.3. Resilience scorecard by ChatGPT

Next, we employed OpenAI's Application Programming Interface (API) and Python programming language to interface with ChatGPT. The API allows users to interact with OpenAI's server programmatically and automate the submission of prompts and retrieval of responses. This process enables the efficient processing of large numbers of tasks that would be impractical via manual interaction. There are three key model specifications required in this process: Prompt design and persona, temperature, and model selection.

Prompts are task instructions given by users to ChatGPT, including specific task descriptions, evaluation criteria, expected outputs and their format. A part of this prompt design is a persona, which is a 'role' given to ChatGPT that sets such characteristics as personalities, tone, and expertise. As the default persona, we used "You are a helpful assistant", the pre-populated persona by OpenAI for their API service (the exact prompts are included in Appendix A and B). Temperature is a parameter that controls the randomness of ChatGPT's output, with lower values generating less random outputs (OpenAI, n.d.-a). Following the approach used in Fu, Wang, and Li

(2024), we set the temperature to 0 for all queries to ensure the consistency and reproducibility as much as possible. By minimizing randomness, we ensure that the model behaves as deterministically as possible – meaning that if different planners or researchers were to input the same policy at different times, they would receive as consistent and reproducible outputs as possible. Regarding model selection, OpenAI provides a range of models tailored to different purposes. For this analysis, we utilized gpt-4o-2024-08-06, OpenAI's most advanced model available at the time of the analysis.

3.3.1. *Task 1 by ChatGPT: selecting relevant policies*

For Task 1, which was to identify policies relevant to flood vulnerability from the Plan (see Figure 1), we used the same three criteria that the human examiners used (i.e. Vulnerability, Place, and Tool Criteria). The OpenAI API documentation webpage (OpenAI, n.d.-c) suggests six strategies for writing clear prompts for ChatGPT: Ask the model to adopt a persona; include details in your query; use delimiters to clearly indicate distinct parts of the input; specify the steps required to complete a task; provide examples; and specify the desired length of the output. We applied these tactics as we saw fit to the three selection guidelines to design three prompts, one for each criterion (the exact prompts are included in Appendix A). Using these prompts, we compared the results from the default persona against an alternative persona, "You are an expert assistant in policy evaluation focused on flooding resilience", to see if changes to the persona improved the results.

Each criterion's prompt was tested separately, and ChatGPT's responses were compared with those of the human examiners, as shown in Table 2. To quantify the level of agreement, we report the following performance indicators:

$$\%\ \text{Agreement} : \frac{true\ positive + true\ negative}{true\ positive + true\ negative + false\ positive + false\ negative}$$

$$\text{Recall} : \frac{true\ positive}{true\ positive + false\ negative}$$

$$\text{Precision} : \frac{true\ positive}{true\ positive + false\ positive}$$

$$\text{Specificity} : \frac{true\ negative}{true\ negative + false\ positive}$$

$$\text{F1 Score} : \frac{2 * Precision * Recall}{Precision + Recall}$$

$$\text{Cohen's Kappa} : \frac{p_0 - p_e}{1 - p_e}$$

where p_0 is the proportion of observed agreements and p_e is the proportion of agreements expected by chance (Sim and Wright 2005, 258).

Percent agreement is "Of all policies, what proportion did the human examiners and ChatGPT assign the same label?" Recall measures "Of all policies labeled as relevant by the human examiners, what proportion did ChatGPT correctly identify as relevant?" This measures the model's ability to identify all relevant policies in a given plan without omission. Precision measures "Of all policies deemed relevant by ChatGPT, what proportion are actually relevant based on human examination?" and

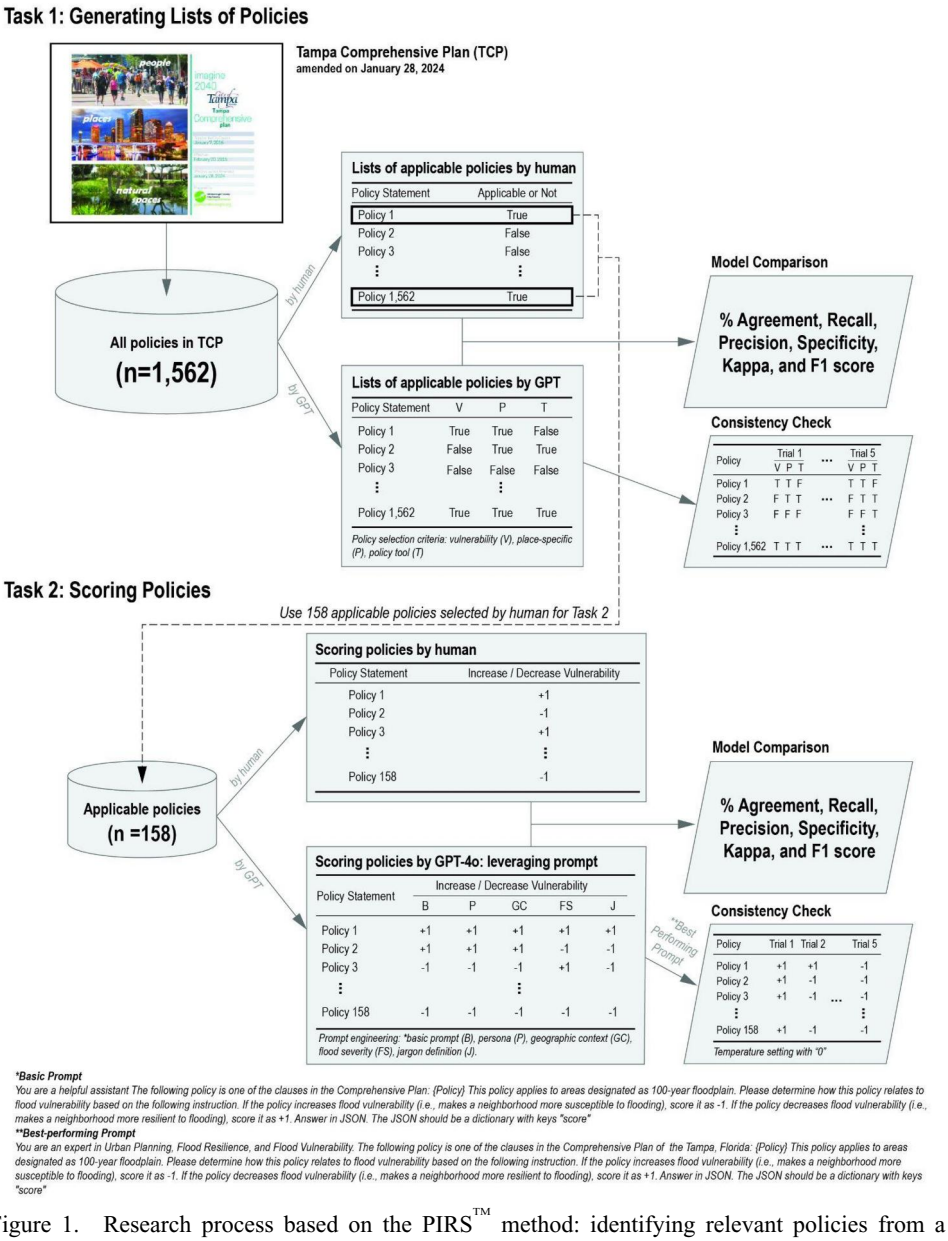


Figure 1. Research process based on the PIRS™ method: identifying relevant policies from a plan (Task 1) and scoring the relevant policies (Task 2).

therefore is about the trustworthiness of ChatGPT's selection of relevant policies. High precision implies efficiency, as it minimizes the time human planners must spend filtering out incorrectly flagged policies. Specificity focuses on the ability to identify irrelevant policies (i.e. false cases) and measures "Of all policies that are deemed irrelevant by the human examiners, what proportion did ChatGPT correctly label as irrelevant?" Given that the vast majority of a comprehensive plan's text may be unrelated to flooding, specificity measures how successfully the model filters out this 'noise' and correctly classifies irrelevant policies as such. F1 score is a harmonic mean

Table 2. Example of a comparison table for Task 1 - Selection of relevant policies, juxtaposing the results by the human examiner and ChatGPT.

		ChatGPT's Result (true = relevant)				
Policy	Human Examiner's Result (true = relevant)	Vulnerability Criterion	Place Criterion	Tool Criterion	Combined Criterion (AND) (true if all 3 criteria are true)	Combined Criterion (OR) (true if one or more of 3 criteria are true)
Policy 1	true	true	true	false	false (2 trues)	true (2 trues)
Policy 2	false	false	false	false	false (0 trues)	false (0 trues)
…	…	…	…	…	…	…
Policy 1562	false	true	true	true	true (3 trues)	true (3 trues)

of the precision and recall (Chicco and Jurman 2020). Kappa statistic, which measures "the proportion of agreement beyond that expected by chance" (Sim and Wright 2005, 258), was included because percent agreement can be inflated by chance if, for example, 90% of policies are relevant (or irrelevant). In such cases, if ChatGPT blindly labels all policies as relevant (or irrelevant), the percent accuracy would be 90% just by chance.

We also investigated the consistency of ChatGPT's evaluation of a given policy by giving the same task to ChatGPT multiple times. Following the recommendation for consistency evaluation for LLMs in Lee, Hong, and Thorne (2025), we had ChatGPT repeat Task 1 five times to evaluate whether its responses changed. This step addresses a well-documented limitation of LLMs, namely their non-deterministic nature (Atkins *et al*. 2024).

3.3.2. Task 2 by ChatGPT: scoring the selected policies as increasing/decreasing vulnerability

The detailed analytical procedures for Task 2 are presented in Figure 1. Specifically, each policy was evaluated using the basic prompt as well as four engineered prompts that incorporated various contextual information described below (Table 3). To isolate the effect of specific contextual information, each engineering element shown below was added individually rather than cumulatively to the basic prompt.

Table 3. Example of a comparison table for Task 2 - Evaluation of policies, juxtaposing the results by the human examiner and ChatGPT.

		ChatGPT's Result (+1: decrease vulnerability; −1: increase vulnerability)				
Policy	Human Examiner's Result	Basic Prompt	Expert Persona	Geographic Context	Flood Severity	Jargon Definition
Policy 1	+1	+1	+1	+1	+1	+1
Policy 2	−1	+1	−1	−1	+1	−1
…	…					
Policy 158	+1	+1	+1	+1	+1	+1

- Expert Persona: Instead of the default persona in the basic prompt, we assigned the model a specific role: "You are an expert assistant in policy evaluation focused on flooding resilience." This aimed to shift the model's reasoning depth from generic conversation to domain-specific professional judgement.
- Geographic Context: It is possible that the same policy can have different impacts depending on geographic locations. We explicitly incorporated 'Tampa' to assess whether ChatGPT could incorporate geographical contexts specific to Tampa into its evaluations.
- Flood Severity: Given that Tampa is projected to face increasing flood risks due to changing climate conditions, including extreme rainfall and sea level rise, the engineered prompts were designed to evaluate the impact of policies under the assumption of intensified flood severity.
- Jargon Definition: Planning documents can be dense with acronyms and technical terms. Following Fu, Wang, and Li (2024), we provided explicit definitions of planning jargon to prevent misinterpretation and enhance the accuracy of the evaluation.

Detailed descriptions of the basic prompt and engineered prompts are provided in Appendix B.

Unlike Task 1, where there was a natural positive class (i.e. relevant policies), Task 2 involves two equally meaningful categories (+1 and −1). To avoid arbitrarily designating one direction as the positive class, we report macro-averaged versions of the performance indicators, which compute performance indicators for each category independently and then average them.

Finally, as in Task 1, we examined whether ChatGPT generated consistent results across iterations using the best-performing prompt developed above. Following Lee, Hong, and Thorne (2025), we requested ChatGPT to evaluate the policies five times and analyzed whether consistent results were produced across these iterations.

4. Findings

4.1. Task 1: selecting policies relevant to flood vulnerability

4.1.1. Performance evaluation

The Plan contained 1,562 policies or policy-like statements, with 158 identified by the human examiners as relevant to flood vulnerability. Table 4 shows how well ChatGPT's results aligned with those of the human examiners. The highest percent agreement was found for Combined (AND) Criteria, which classified a policy as relevant to flood resilience only if ChatGPT identified it as relevant under all three criteria of Vulnerability, Place, and Tool. This resulted in about 90% alignment between human and ChatGPT labels. Similar agreement levels were observed for the Vulnerability and Tool Criteria. Criteria with high percent agreement also tended to exhibit high specificity, with the Combined (AND) Criteria achieving the highest specificity of 97%, meaning ChatGPT correctly identified 97% of irrelevant policies. However, because almost 90% of all policies were deemed by the human examiners as irrelevant, high percent agreement or high specificity alone does not necessarily indicate ChatGPT's good performance. A classifier that predicts all policies to be irrelevant, for example, would also be correct nearly 90% of the time[1].

Table 4. Performance evaluation metrics for different evaluation criteria (default persona).

	Vulnerability Criterion	Place Criterion	Tool Criterion	Combined (AND) (true if all 3 criteria are true)	Combined (OR) (true if one or more of 3 criteria are true)
% Agreement	0.852	0.537	0.825	0.898	0.473
Recall	0.449	0.778	0.595	0.285	0.924
Precision	0.329	0.152	0.310	0.500	0.153
Specificity	0.897	0.510	0.851	0.968	0.422
Kappa	0.298	0.102	0.317	0.312	0.107
F1 Score	0.380	0.254	0.408	0.363	0.263

Note. Cohen suggested the Kappa result be interpreted as follows: values ≤ 0 as indicating no agreement and $0.01 \sim 0.20$ as none to slight, $0.21 \sim 0.40$ as fair, $0.41 \sim 0.60$ as moderate, $0.61 \sim 0.80$ as substantial, and $0.81 \sim 1.00$ as almost perfect agreement (McHugh 2012).

The highest recall of 0.924 was found for Combined (OR) Criterion, indicating that of the 158 relevant policies, ChatGPT correctly labeled 92% of them as relevant (146 true positives). This high recall, however, was accompanied by a low precision of 0.153, a phenomenon commonly known as the precision-recall trade-off. This trade-off refers to the empirical tendency for recall and precision to be inversely related (Buckland and Gey 1994). The 0.153 precision indicates that, of all policies that ChatGPT labeled as relevant, 15.3% were marked as relevant by human examiners. The precision-recall relationship suggests that while the Combined (OR) Criterion produces a high number of true positives, it also results in a substantial number of false positives (807 false positives in this case). This trade-off was evident across various evaluation criteria.

Evaluating 1,562 policies using ChatGPT based on the Vulnerability Criterion took approximately 99 to 107 min (around 4.11 s per policy) and cost $4 to $5 USD. The evaluation times and costs for the Place and Tool Criteria were similar (accessed January 16, 2025). Since the Combined Criteria included all three individual criteria, the total cost was roughly three times higher. In contrast, human examiners spent a considerably longer time. While it is difficult to accurately estimate the total time because human examiners worked intermittently over several days, our rough estimate, based on the duration per policy, is over 100 h per examiner for 1,562 policies. Combined, the two examiners spent over 200 h on this task. Note that the time reported for ChatGPT reflects only the API processing time and does not include the time invested in prompt design, coding, or post-hoc result inspection. Similarly, the human examiner's time only includes time spent on evaluating the policies and does not include prior training or familiarization with the evaluation framework.

In summary, the Combined (AND) and Combined (OR) Criteria outperformed other single-criterion approaches. The Combined (AND) Criteria, which labeled policies to be relevant only when all three of the Vulnerability, Place, and Tool Criteria viewed them as relevant and therefore the most demanding criteria, showed the highest % agreement (0.898) and specificity (0.968) but the lowest recall (0.285). Combined (OR) Criteria, which viewed policies to be relevant if any one of Vulnerability, Place, or Tool Criteria viewed them relevant and therefore the most permissive, showed the highest recall (0.924) but the second lowest precision (0.153) and the lowest specificity (0.422).

Table 5. Performance evaluation metrics for different evaluation criteria (alternative persona).

	Vulnerability Criterion	Place Criterion	Tool Criterion	Combined (AND) (true if all 3 criteria are true)	Combined (OR) (true if one or more of 3 criteria are true)
% Agreement	0.846	0.511	0.831	0.900	0.446
Recall	0.576	0.766	0.627	0.367	0.918
Precision	0.345	0.143	0.326	0.509	0.146
Specificity	0.877	0.482	0.854	0.960	0.393
Kappa	0.349	0.084	0.341	0.373	0.093
F1 Score	0.431	0.241	0.429	0.426	0.252

4.1.2. Prompt engineering (persona)

The result reported above used the default persona (i.e. You are a helpful assistant). We examined whether an alternative persona (i.e. You are an expert assistant in policy evaluation focused on flooding resilience) can meaningfully change ChatGPT's performance. Overall, the performance change between the default and alternative persona was not substantial, with mixed improvements and declines across different criteria and evaluation metrics (See Table 5). The recall of Vulnerability showed the most noticeable improvement (0.449 → 0.576; difference of 0.127), followed by the recall of Combined Criteria (AND) (0.285 → 0.367; difference of 0.082).

4.1.3. Consistency evaluation

As outlined in Section 3.3.1, we queried ChatGPT five times for each of the criteria – Vulnerability, Place, and Tool – across 1,562 policies to assess response consistency. For the Vulnerability Criterion, ChatGPT returned consistent responses across five repeats for 1,501 out of 1,562 policies (96.1%). Place Criteria had the lowest, but still reasonably high, consistency, with 1,409 out of 1,562 policies yielding consistent responses across five repeats (90.2%). The Tool Criteria had more consistent responses: 1,469 out of 1,562 policies (94.0%) yielded consistent responses over five iterations.

4.2. Task 2: evaluating whether the selected policies increase or decrease flood vulnerability

4.2.1. Performance evaluation

The results of Task 2 are presented in Table 6. Notably, even without prompt engineering, the basic prompt achieved an agreement rate of nearly 90% with human examiners' evaluations. Specifically, out of the 158 policies assessed, 142 matched the evaluations of human examiners, with discrepancies observed in only 16 policies.

Notably, the differences in accuracy between various prompt design strategies (e.g. Persona) and the basic prompt were minimal. While certain performance indicators showed slight improvements with specific prompt designs (e.g. Persona achieved the highest macro precision at 0.915), the overall results were mixed, with marginal

Table 6. Performance evaluation metrics for Task 2.

	Basic Prompt	Expert Persona	Geographic Context	Flood Severity	Jargon Definition
% Agreement	0.899	0.905	0.899	0.886	0.886
Macro Recall	0.880	0.886	0.881	0.866	0.866
Macro Precision	0.900	0.915	0.895	0.891	0.891
Kappa	0.777	0.794	0.775	0.752	0.752
Macro F1 Score	0.888	0.897	0.887	0.876	0.876

differences across other metrics like macro recall, macro specificity, and macro F1 score. Prompt engineering is widely regarded as an effective strategy for enhancing the performance of LLMs (e.g. Polak and Morgan 2024). However, our findings deviate from this general assumption, aligning with a few prior studies suggesting that prompt engineering does not consistently improve accuracy across all tasks (Gan and Mori 2023; Shin *et al*. 2023; Webson and Pavlick 2022). Specifically, although Persona performed slightly better in precision (0.915) and kappa (0.794), these gains were not substantial enough to indicate a significant overall advantage. Furthermore, U. Lee *et al*. (2024) observed that while prompt engineering may enhance model performance for certain tasks, it often fails to significantly improve outcomes when tasks involve complex sentence structures or nuanced contextual understanding.

In terms of evaluation efficiency, the ChatGPT model completed the task of scoring 158 policies as $+1$ or -1 in a total of 288 s (4.8 min), averaging just 1.82 s per policy. It cost about $0.19 USD to evaluate 158 policies (about $0.0012 USD per policy). In contrast, on average each human examiner – whose scoring times varied depending on the complexity and ambiguity of each policy – required a total of 36,340 s (606 min), averaging 230 s per policy. Combined, the two examiners spent over 20 h on this task. This comparison underscores the remarkable efficiency of ChatGPT in plan evaluation, delivering comparable accuracy within a significantly shorter timeframe.

4.2.2. Consistency analysis

To address the non-deterministic nature of ChatGPT, we examined the consistency of its responses for the 158 relevant policies across five iterations. Specifically, each policy was assessed five times using the same prompt (i.e. under identical conditions), and if ChatGPT produced identical responses across all iterations, it was considered capable of providing consistent evaluations for the policies. This analysis was conducted based on the basic prompt. The analysis revealed that out of 158 policies, ChatGPT generated completely identical responses in 156 cases (98.73%) across the five iterations.

5. Discussion and conclusions

This study explored the potential of ChatGPT in conducting policy evaluation using the Plan Integration for Resilience Scorecard™ method by comparing its evaluation results with those of trained human examiners. Using the Imagine

2040: Tampa Comprehensive Plan as a case study, the study focused on two critical tasks: (1) identifying policies relevant to flood vulnerability from 1,562 policies and (2) evaluating whether the selected 158 policies increase or decrease flood vulnerability.

5.1. Summary of findings

For Task 1, ChatGPT showed mixed performance in identifying policies relevant to flood vulnerability across various criteria. While % Agreements were generally high, it was likely due to the imbalance in data, and the trade-off between recall and precision was evident. This trade-off led to varying F1 scores, ranging from 0.241 to 0.431, underscoring the need for careful selection of evaluation criteria. The modified persona used in this study did not seem to provide meaningful improvements. The consistency of ChatGPT's responses was high for all three criteria, ranging between 90% and 96%. Although non-determinism is one of ChatGPT's well-documented inherent issues (Atkins *et al*. 2024), an appropriate temperature parameter (temperature = 0) and prompt design seem to be effective in controlling the model's randomness and ensuring greater reliability.

Task 2 showed markedly better performance, with ChatGPT achieving high scores across evaluation metrics. Most scores approached 0.9, and the Kappa statistic of 0.777 indicated substantial agreement, approaching the threshold for 'almost perfect' agreement (McHugh 2012). The prompt engineering used in this study did not result in substantial improvements over the base prompt, suggesting that ChatGPT delivers robust performance with its baseline capabilities (Table 6). Additionally, the tool demonstrated high consistency, with 99% of the policies yielding the same responses across repeated queries. These findings suggest that ChatGPT's non-deterministic behavior is less pronounced than often assumed, further supporting its reliability for plan evaluation.

When it comes to time and costs associated with performing each task, ChatGPT demonstrated remarkable efficiency compared to human examiners. For Task 1, ChatGPT evaluated all 1,562 policies in the Tampa Comprehensive Plan in approximately 100 min at a cost of about $4 to $5 USD. In contrast, human examiners required over 6,000 min per examiner to complete the same task, making ChatGPT over 60 times faster. For Task 2, ChatGPT completed the evaluation in just 4.8 min at a cost of approximately $0.2 USD, whereas human examiners required an estimated 606 min, demonstrating that ChatGPT was more than 120 times faster. These findings highlight ChatGPT's potential as an effective tool for plan evaluation, enabling planners to quickly analyze extensive datasets with minimal time and financial investment, thereby enhancing cost-effectiveness.

5.2. Implications to practitioners

These findings have important implications for practitioners and researchers, as the ongoing challenges of climate change create significant uncertainty and reshape the social, environmental, and political contexts that influence urban planning and policy-making. In response to the increasing risks posed by natural hazards and the growing challenges in addressing them, cities must urgently evaluate and adapt their existing

plans and policies, with the possibility of conducting frequent reassessments to keep pace with the evolving conditions.

Under such circumstances, ChatGPT's ability to reduce monetary, temporal, and human resources demands can make timely plan evaluation possible. Depending on the urgency and complexity of the plan, such rapid initial evaluation results by ChatGPT can either directly inform policy revision or serve as inputs for a more robust and gradual plan evaluation process involving public officials, private sectors, and community and non-profit organizations.

Moreover, the finding that ChatGPT performed well out of the box suggests that it can be an accessible and easy-to-use tool for practicing planners and policymakers, requiring minimal additional efforts for prompt engineering. This reduces the need for extensive customization and enhances the method's accessibility for practitioners. With basic programming proficiency, planning departments can adopt this automated evaluation method seamlessly. Code templates in Python and R used in this study can be provided upon request, facilitating broader adoption.

The growing use of AI-based tools such as ChatGPT raises important questions regarding the future role of planning professionals. While the automation of early-stage policy screening and evaluation may reduce the need for manual, labor-intensive review tasks, existing research suggests that such technologies are more likely to reconfigure professional roles rather than reduce overall demand for professionals (Autor 2015; Brynjolfsson, Mitchell, and Rock 2018). The role of planners is expected to shift toward higher-order activities, including interpreting policy implications, exercising professional judgment, integrating local and contextual knowledge, and facilitating stakeholder deliberation. ChatGPT should therefore be understood as a decision-support tool that redirects professional effort rather than replaces professional expertise, particularly in complex and value-laden planning contexts.

Empirical evidence from American Planning Association (APA) practitioner surveys further underscores the limits of AI substitution in planning practice. A survey on the use of AI found that only about 7% of respondents (or their organizations) reported that they plan to adopt AI tools in the near future, despite growing public attention and technological advances (Sanchez *et al*. 2023). When asked which planning tasks are least suitable for AI, interestingly, plan review was one of the few areas that received high scores for both suitability and unsuitability, indicating ambivalence among practitioners. While AI may support efficiency in text-based tasks, its role in interpretive, value-laden, and politically embedded processes remains contested (Sanchez *et al*. 2023). Even as AI capabilities improve, institutional adoption does not always follow, and professional acceptance may lag behind technical feasibility.

Relatedly, the increasing reliance on data-driven AI tools may incentivize professionals or institutions to guard proprietary datasets, localized knowledge, or interpretive frameworks in order to retain professional relevance. Prior studies on data-driven governance caution that such tendencies toward data enclosure and opacity may emerge alongside algorithmic decision support (Janssen and Kuk 2016). However, in planning practice, these dynamics are constrained by institutional norms emphasizing transparency, accountability, and collaborative decision-making. Even when policy evaluation processes are partially automated, the processes of negotiation, value-based judgment, and consensus-building remain fundamentally human responsibilities, consistent with long-standing planning theory that characterizes planning as an inherently normative and political process (Flyvbjerg 1998; Rittel and Webber 1973).

From the application perspective, one of the main caveats identified in this study for users is that the trade-off between recall and precision of Task 1 indicates that users should carefully select evaluation criteria that fits their purpose when using ChatGPT to identify relevant policies from plans. A criterion with high recall ensures most relevant policies are identified but may result in many false positives, necessitating human refinement. When there is a substantial number of policies to be reviewed or in communities with limited resources, this high number of false positives may be taxing. High precision reduces false positives but risks omitting relevant policies. Therefore, selecting the right balance depends on the policy evaluator's purpose. In this case, ChatGPT can serve as a supplementary member of a larger evaluation team.

However, this trade-off concern may only be relevant for general plans that include policies unrelated to resilience – for example, the comprehensive plan examined in this study. In contrast, when evaluated plans consist solely of resilience-related policies, as seen in climate action or dedicated resilience plans, policy examiners can proceed directly to Task 2 without concerns for this trade-off. As shown in Table 6, both recall and precision are high when assessing whether the selected policies increase or decrease flood vulnerability.

Finally, the consistent results observed across repeated queries in both Task 1 and Task 2 highlight ChatGPT's potential to improve the reliability of policy evaluations in practical applications. Our findings, with a consistency rate of 90 to 99%, demonstrate that the non-determinism issue – where variability in evaluation results could pose significant challenges to reliability in practical use – can be effectively mitigated. This high level of consistency reinforces ChatGPT's applicability as a reliable and stable tool for policy evaluation, offering practitioners greater confidence in its suitability for real-world applications.

It is important to reiterate that this is an exploratory study, aimed at providing an early assessment of the potential for future application and at laying the foundation for future research. As LLMs continue to advance rapidly, the strengths identified in this study are likely to become even more pronounced, while the limitations are expected to diminish over time. Future studies will benefit from closely monitoring AI developments and adapting research directions accordingly.

5.3. Limitation

While the findings of this research offer valuable insights into the use of AI in plan evaluation, certain limitations must be acknowledged. First, this study is confined to a single comprehensive plan, which limits the ability to generalize findings across the wide variety of plan qualities and structures. While this focus on a single document enabled a granular analysis that breaks the process of plan evaluation into steps and specific criteria, it limits generalizability to other types of plans and cities. Future research can broaden the scope of plans and policies to, for example, hazard mitigation plans, transportation plans, or housing plans to understand whether the findings from this study apply to the complexities and languages used in other planning contexts.

In this regard, prior work suggests that LLM performance can be highly context- and task-specific (Oh and Demberg 2025; Zheng *et al*. 2025). Future research should therefore examine whether the comparatively strong performance observed here, in a relatively tangible domain centered on flood resilience and safety, generalizes to plans

that hinge more on aesthetic and social values. Such evaluations often require more intangible and nuanced judgment and may be more challenging for ChatGPT.

Second, ChatGPT's reliance on its training data to interpret and evaluate policies limits its ability to capture localized nuances, technical language, or culturally specific terms embedded in regional planning documents, which can result in evaluation inaccuracies (J. Kim, Lee, *et al*. 2024). The resulting scores from AI must be carefully reviewed by human experts, and professional judgment remains indispensable.

Third, because this study focused on Tampa's comprehensive plan, it remains unclear whether the model's performance generalizes to other cities, regions, or countries, where policy language, structure, cultural background, and priorities may differ significantly. Future research should test the method on plans from diverse cities, regions, and institutional contexts. In doing so, future research can aim to improve the model's applicability across different contexts by exploring (1) more advanced prompt engineering, following the development of the field of prompt engineering and (2) more refined and detailed definition of resilience and flood vulnerability as they apply to diverse policy contexts and urban settings. In their documentation for ChatGPT, OpenAI writes, "To use the [OpenAI's text generation] models effectively, we include instructions and sometimes several examples in a prompt. Using demonstrations to show how to perform a task is often called few-shot learning" (OpenAI n.d.-b). Few-shot learning can provide ChatGPT more information to generate responses that better align with the judgements of local planners and policymakers. Additionally, integrating the reasoning process of a specific task into the prompt, known as the Chain of Thought Prompting strategy (Wei *et al*. 2023), is worth exploring to enhance ChatGPT's ability to produce more accurate and context-aware evaluations.

Fourth, another important limitation of this study concerns the nature of professional judgement in planning and its relationship to tacit knowledge (Nonaka and Takeuchi 1995). Professional expertise in urban planning is not developed solely through exposure to texts, but also through internships, mentorship, and practice-based experience. Following Nonaka and Takeuchi's knowledge creation theory, such experiential judgement can be understood as *tacit knowledge*, which is a context-sensitive, practice-embedded understanding that is difficult to fully articulate or formalize (Nonaka and Takeuchi 1995). As highlighted in research on tacit knowledge transfer in urban design (Aranda-Mena 2018), such knowledge often requires deliberate elicitation to be externalized and shared. Because LLMs are trained primarily on textual data, their access to this form of knowledge is necessarily indirect and incomplete.

At the same time, it is important to recognize that contemporary planning practice increasingly involves efforts to externalize tacit professional reasoning into more explicit forms, such as structured evaluation frameworks, resilience scorecards, professional guidelines, case documentation, and community reports. To the extent that experiential knowledge becomes codified and systematically documented, AI systems may approximate certain aspects of professional judgement, particularly in rule-based and text-driven evaluation tasks. However, such externalization remains partial, and context-sensitive interpretation, ethical judgement, and place-specific reasoning continue to require human oversight. Accordingly, AI-based plan evaluation should be understood as a complementary efficiency-enhancing tool rather than a substitute for professional expertise.

Fifth, this study focused only on evaluating the default, off-the-shelf model whereas OpenAI offers methods to fine-tune their models. OpenAI further states, "Fine-tuning improves on few-shot learning by training on many more examples than

can fit in the prompt, letting you achieve better results on a wide number of tasks. Once a model has been fine-tuned, you won't need to provide as many examples in the prompt". Unlike prompt engineering, which seeks to improve outputs without modifying the model, fine-tuning adjusts the model itself to better align with domain-specific tasks. Fine-tuning allows municipalities to develop a customized version of ChatGPT that incorporates insights from local planners and policymakers, thereby improving its performance in interpreting and evaluating regional planning documents.

5.4. Conclusions

This study highlighted ChatGPT's potential as an innovative tool for flood resilience plan evaluation. Its efficiency in processing large volumes of data at low cost and time makes it particularly valuable for resource-constrained municipalities (Davis 2022; King 2023) and planning departments or those faced with rapidly shifting social and environmental contexts. By helping human examiners shortlist relevant policies in Task 1 or potentially acting as an independent evaluator in Task 2, ChatGPT can significantly reduce the time and effort required for plan evaluations. While human oversight remains essential for interpreting nuanced policy contexts and refining incorrect predictions, ChatGPT's ability to handle large volumes of textual data and provide consistent outputs underscores its potential as a promising solution for rapid and efficient plan evaluation, helping communities become better prepared for the increasing risks posed by climate change.

Note

1. We therefore report Cohen's Kappa statistic, which corrects for this chance agreement by taking expected agreement into account, to address this imbalance, thereby providing a more robust metric.

Declaration of generative AI in scientific writing

During the preparation of this work the author(s) used ChatGPT to (1) assess its capacity for plan evaluation and (2) improve the readability and language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This research was supported by the Yonsei University Research Fund of 2025-22-0145.

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Appendices

Appendix A

We provide the prompts used for Task 1, as explained in Section 3.3.1 of the manuscript for the method and in Section 4.1 for the results. The parts in the prompts wrapped with {} are automatically replaced with other appropriate texts using the Python programming language.

A1. Prompt for ChatGPT to identify policies that can affect vulnerability to flooding (Vulnerability Criterion)

Prompt:

"The following policy is one of the clauses in the Comprehensive Plan: **{policy}**. Assume this policy applies to areas prone to flooding. If the policy either increases or decreases flood vulnerability (i.e., makes a neighborhood more or less susceptible to flooding), score it as 'true'. If the policy does not relate to any changes in flood vulnerability, score it as 'false'. In two to three sentences, explain your rationale for your score, detailing how the policy impacts flood vulnerability, if applicable."

Note that **{policy}** is replaced with the actual policy being evaluated in each iteration.

A2. Prompt for ChatGPT to identify policies that contain place-specific term(s) that can be mapped (Place Criterion)

Prompt:

"The following policy is one of the clauses in the Comprehensive Plan: **{policy}**. Assume this policy applies to areas prone to flooding. If the policy includes at least one mappable or place-specific term, score it as 'true'. If the policy does not contain at least one mappable or place-specific term, score it as 'false'. In two to three sentences, explain your rationale for your score, specifying what terms were present and how this affected your assessment."

Note that **{policy}** is replaced with the actual policy being evaluated in each iteration.

A3. Prompt for ChatGPT to identify policies that contain specific policy tools (Tool Criterion)

Prompt:

"The following policy is one of the clauses in the Comprehensive Plan: **{policy}**. Assume this policy applies to areas prone to flooding. Please reference the list of recognizable policy tools wrapped by *** ***: **{policy_tools}**. If the policy includes at least one of these policy tools, score it as 'true'. If the policy does not include any of these policy tools, score it as 'false'. In two to three sentences, explain your rationale for your score, specifying which tools were present or absent and how this affected your assessment."

Note that **{policy}** is replaced with the actual policy being evaluated in each iteration. Bolding of the text in the prompts was not applied in the original prompt but is here for better legibility. Similarly, **{policy_tools}** is replaced with the following list and description of policy tools in the quotation marks, which are taken from the Guidebook

> "**Permitted Land Use** - Provision regulating the types of land use (e.g. residential, commercial, industrial, open space, etc.) permitted in areas of community; may be tied to zoning code., **Density of Land Use** - Provision regulating density (e.g. units per acre); may be tied to zoning code., **Subdivision Regulations** - Provision controlling the subdivision of parcels into developable units and governing the design of new development (e.g. site storm water management)., **Zoning Overlays** - Provision to use zoning overlays that restrict permitted land use/density in hazardous areas; may be special hazard zones or sensitive open space protection zones., **Setbacks or Buffer Zones** - Provision requiring setbacks or buffers around hazardous areas (e.g. riparian buffers and ocean setbacks)., **Cluster Development** - Provision requiring clustering of development away from hazardous areas, such as through conservation subdivisions., **Acquire Land & Property** - Purchase land/property in hazard area., **Open Space or Easement Requirement/Purchase** - Provision encouraging open space purchase by the community or open space easements as an element of development approval., **Transfer/ Purchase of Development Rights** - Provision for transferring development rights to control density; may be transfer of development rights or purchase of development rights. **Density Bonuses** - Density bonuses such as ability to develop with greater density in return for dedication or donation of land in areas subject to hazards., **Tax Abatement** - Tax breaks offered to property owners and developers who use mitigation methods for new development., **Impact/Special Study/Protection Fees** - Provision requiring impact fees, special study fees, or protection fees for development in hazardous areas; fees could cover costs of structural protection., **Land Suitability** - Hazards are one of the criteria used in analyzing and determining the suitability of land for development., **Site Review** - Provision requiring addressing hazard mitigation in process of reviewing site proposals for development., **Design/Construction Guidelines/ Requirements** - Guidelines or requirements that apply to the design or construction of developments in hazard areas., **Siting** - Provision to site public facilities, including municipal buildings and public housing, out of hazard areas., **Sizing/Capacity** - Provision limiting capacity of public facilities, including public housing, in hazard areas to cap amount of development., **Development Moratorium** - Provision imposing a moratorium on development for a set period of time after a hazard event to allow for consideration of land use change., **Post-Disaster Land Use Change** - Provision related to changing land use regulations following a hazard event; may include redefining allowable land uses after a hazard event., **Post-Disaster Capital Improvements** - Provision related to adjusting capital improvements to public facilities following a hazard event., **Infrastructure "Hardening" or Weatherproofing** - Provision encouraging or requiring development in hazard zones to increase structural resilience to hazards., **Elevating** - Provision pertaining to the physical elevation of structures in hazard zones., **Drainage Improvements or Flood Control** - Provision that pertains to drainage or flooding issues within the community., **Ecosystem Enhancement** - Provision that seeks to improve or preserve the functioning of the natural environment within the community., **Slope/Dune Stabilization** - Provision that pertains specifically to stabilization of slopes or dunes or seeks to control erosion."

Appendix B

We provide the prompts used for Task 2, as explained in Section 3.2.2 of the manuscript for the method and in Section 4.2 for the results.

B1. Basic prompt of ChatGPT for Task 2

Prompt:

"The following policy is one of the clauses in the Comprehensive Plan: **{Policy}** This policy applies to areas prone to flooding. Please determine how this policy relates to flood vulnerability based on the following instruction. If the policy increases flood vulnerability (i.e., makes a neighborhood more susceptible to flooding), score it as -1. If the policy decreases flood vulnerability (i.e., makes a neighborhood more resilient to flooding), score it as $+1$. In two to three sentences, explain why the policy increases or decreases flood resilience. It must be evaluated as either $+1$ or -1.

Answer in JSON. The JSON should be a dictionary with keys "score"'

B2. Engineered prompt of ChatGPT for Task 2

Prompt:

"The following policy is one of the clauses in the Comprehensive Plan of Tampa, Florida, USA: **{Policy}**

This policy applies to areas where flooding is expected to occur. Please determine how this policy relates to flood vulnerability based on the following instruction. If the policy increases flood vulnerability (i.e., makes a neighborhood more susceptible to flooding), score it as -1. If the policy decreases flood vulnerability (i.e., makes a neighborhood more resilient to flooding), score it as $+1$. In two to three sentences, explain why the policy increases or decreases flood resilience. It must be evaluated as either $+1$ or -1.

In evaluating this policy, you may refer to the following definitions of abbreviations- CBD: Central Business District, CM: Coastal Management, ENV: Environment, EPC: Environmental Protection Commission, FDEP: Florida Department of Environmental Protection, FEMA: Federal Emergency Management Agency, FLUM: Future Land Use Map, HSG: Housing, LU: Land Use, MPO: Metropolitan Planning Organization, ROS: Recreation and Open Space, SWRPA: special water resource protection area, TOD: transit oriented development, SWFWMD: Southwest Florida Water Management District, WRPAs: Wellhead Resource Protection Areas, MAP: Municipal Airport Compatibility, FAP: Flight Accident Potential, FLUE: Future Land Use Element, LPD: Local Planning Districts, FDOT: Florida Department of Transportation, DPW: Department of Public Works, SWM: Storm Water Management.

Answer in JSON. The JSON should be a dictionary with keys "score"'

Note: The underlined text indicates the information added during prompt engineering.